

Hands on 1 Spring Data JPA - Quick Example:

SQL Queries:

-- Create Schema

CREATE SCHEMA ormlearn;

USE ormlearn;

-- Create Table with Primary Key constraint

```
CREATE TABLE country (  
    co_code VARCHAR(2) PRIMARY KEY,  
    co_name VARCHAR(50)  
);
```

-- Insert Sample Records for testing

INSERT INTO country VALUES ('IN', 'India');

INSERT INTO country VALUES ('US', 'United States of America');

application.properties:

Spring Framework and Application Logging

logging.level.org.springframework=info

logging.level.com.cognizant=debug

Hibernate Core SQL Trailing Logs

logging.level.org.hibernate.SQL=trace

logging.level.org.hibernate.type.descriptor.sql=trace

Custom Console Log Formatting Pattern

logging.pattern.console=%d{dd-MM-yy} %d{HH:mm:ss.SSS} %-20.20thread %5p %-25.25logger{25} %25M %4L %m%n

Database Connectivity Configuration

spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver

spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn

spring.datasource.username=root

spring.datasource.password=YOUR_PASSWORD_HERE

JPA / Hibernate Lifecycle Management

spring.jpa.hibernate.ddl-auto=validate

Country.java:

```
package com.cognizant.ormlearn.model;
```

```
import jakarta.persistence.Column;
```

```
import jakarta.persistence.Entity;
```

```
import jakarta.persistence.Id;
```

```
import jakarta.persistence.Table;
```

```
@Entity
```

```
@Table(name="country")
```

```
public class Country {
```

```
    @Id
```

```
    @Column(name="co_code")
```

```
    private String code;
```

```
    @Column(name="co_name")
```

```
    private String name;
```

```
    // Getters and Setters
```

```

public String getCode() { return code; }

public void setCode(String code) { this.code = code; }


public String getName() { return name; }

public void setName(String name) { this.name = name; }


@Override

public String toString() {

    return "Country [code=" + code + ", name=" + name + "]";

}

}

```

CountryRepository.java:

```

package com.cognizant.ormlearn.repository;


import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;
import com.cognizant.ormlearn.model.Country;


@Repository

public interface CountryRepository extends JpaRepository<Country, String> {

}

```

CountryService.java:

```

package com.cognizant.ormlearn.service;


import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

```

```
import com.cognizant.ormlearn.model.Country;
import com.cognizant.ormlearn.repository.CountryRepository;
```

```
@Service
```

```
public class CountryService {
```

```
    @Autowired
```

```
    private CountryRepository countryRepository;
```

```
    @Transactional(readOnly = true)
```

```
    public List<Country> getAllCountries() {
```

```
        return countryRepository.findAll();
```

```
    }
```

```
}
```

OrmLearnApplication.java:

```
package com.cognizant.ormlearn;
```

```
import java.util.List;
```

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

```
import org.springframework.boot.SpringApplication;
```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
import org.springframework.context.ApplicationContext;
```

```
import com.cognizant.ormlearn.model.Country;
```

```
import com.cognizant.ormlearn.service.CountryService;
```

```
@SpringBootApplication
```

```
public class OrmLearnApplication {
```

```
private static final Logger LOGGER = LoggerFactory.getLogger(OrmLearnApplication.class);

private static CountryService countryService;

public static void main(String[] args) {
    ApplicationContext context = SpringApplication.run(OrmLearnApplication.class, args);
    LOGGER.info("Inside main");

    countryService = context.getBean(CountryService.class);
    testGetAllCountries();
}

private static void testGetAllCountries() {
    LOGGER.info("Start");
    List<Country> countries = countryService.getAllCountries();
    LOGGER.debug("countries={}", countries);
    LOGGER.info("End");
}
}
```

Output:

```
PS D:\SD\Digital-Nurture-JavaSE-main\local repo\Cognizant-exercises\Week 2\Spring Data JPA> & 'C:\Program Files\Eclipse Adoptium\jdk-25.0.3-hotspot\bin\java.exe' '@C:\Users\adity\AppData\Local\Temp\cp_9u2dhoxpnrwx41mx7aep9nxguwn.argfile' 'com.cognizant.orm_learn.OrmLearnApplication'
```

29-06-26 23:08:05.825	restartedMain	INFO c.c.o.OrmLearnApplication	logStarting	54	Starting OrmLearnApplication using Java 25.0.3 with PID 22624 (D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercises\Week 2\Spring Data JPA\Exercise-1\orm-learn\target\classes started by aditya in D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercises\Week 2\Spring Data JPA)
29-06-26 23:08:05.827	restartedMain	DEBUG c.c.o.OrmLearnApplication	logStarting	55	Running with Spring Boot v4.1.0, Spring v7.0.8
29-06-26 23:08:05.828	restartedMain	INFO c.c.o.OrmLearnApplication	logStartupProfileInfo	652	No active profile set, falling back to 1 default profile: "default"
29-06-26 23:08:05.869	restartedMain	INFO ertyDefaultsPostProcessor	log	69	Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
29-06-26 23:08:06.134	restartedMain	INFO toryConfigurationDelegate	registerRepositoriesIn	145	Bootstrapping Spring Data JPA repositories in DEFAULT mode.
29-06-26 23:08:06.173	restartedMain	INFO toryConfigurationDelegate	registerRepositoriesIn	213	Finished Spring Data repository scanning in 32 ms. Found 1 JPA repository interface.
29-06-26 23:08:06.439	restartedMain	INFO org.hibernate.orm.jpa	logPersistenceUnitInformation	102	HHH008540: Processing PersistenceUnitInfo [name: default]
29-06-26 23:08:06.474	restartedMain	INFO org.hibernate.orm.core	logVersion	41	HHH000001: Hibernate ORM core version 7.4.1.Final
29-06-26 23:08:06.730	restartedMain	INFO SpringPersistenceUnitInfo	addTransformer	129	No LoadTimeWeaver setup: ignoring JPA class transformer
29-06-26 23:08:06.747	restartedMain	INFO c.z.h.HikariDataSource	getConnection	109	HikariPool-1 - Starting...
29-06-26 23:08:06.967	restartedMain	INFO c.z.h.pool.HikariPool	checkFailFast	580	HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@23ef99db
29-06-26 23:08:06.968	restartedMain	INFO c.z.h.HikariDataSource	getConnection	122	HikariPool-1 - Start completed.
29-06-26 23:08:07.033	restartedMain	INFO o.h.o.connections.pooling	logConnectionInfo	179	HHH10001005: Database info: Database JDBC URL [jdbc:mysql://localhost:3306/ormlearn] Database driver: MySQL Connector/J Database dialect: MySQLDialect Database version: 8.0.46 Default catalog/schema: ormlearn/undefined Autocommit mode: undefined/unknown Isolation level: REPEATABLE_READ [default REPEATABLE_READ]

```
PS D:\SD\Digital-Nurture-JavaFSE-main\local_repo\Cognizant-exercises\Week 2\Spring Data JPA> & 'C:\Program Files\Eclipse Adoptium\jdk-25.0.3-hotspot\bin\java.exe' '@C:\Users\adity\AppData\Local\Temp\cp_9udhoxpmrx41mx7aep9nxguwn.argfile' 'com.cognizant.orm_learn.OrmLearnApplication'
```

```
Database JDBC URL [jdbc:mysql://localhost:3306/ormlearn]
Database driver: MySQL Connector/J
Database dialect: MySQLDialect
Database version: 8.0.46
Default catalog/schema: ormlearn/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
```

29-06-26 23:08:07.471	restartedMain e.transaction.jta.platform' to enable JTA platform integration)	INFO org.hibernate.orm.core	initiateService	53	HH#000489: No JTA platform available (set 'hibernat
29-06-26 23:08:07.509	restartedMain instance unit 'default'	INFO rEntityManagerFactoryBean	buildNativeEntityManagerFactory	464	Initialized JPA EntityManagerFactory for pers
29-06-26 23:08:07.564	restartedMain r will be used.	INFO .q.QueryEnhancerFactories	<clinit>	48	Hibernate is in classpath; If applicable, HQL parse
29-06-26 23:08:07.731	restartedMain ss running for 2.431)	INFO c.c.o.OrmLearnApplication	logStarted	60	Started OrmLearnApplication in 2.162 seconds (proce
29-06-26 23:08:07.734	restartedMain	INFO c.c.o.OrmLearnApplication	main	20	Inside main
29-06-26 23:08:07.734	restartedMain	INFO c.c.o.OrmLearnApplication	testGetAllCountries	27	Start
29-06-26 23:08:07.841	restartedMain	DEBUG org.hibernate.SQL	logStatement	126	select c1_0.co_code,c1_0.co_name from country c1_0
29-06-26 23:08:07.867	restartedMain code=US, name=United States of America]]	DEBUG c.c.o.OrmLearnApplication	testGetAllCountries	29	countries=[country [code=IN, name=India], country [
29-06-26 23:08:07.868	restartedMain	INFO c.c.o.OrmLearnApplication	testGetAllCountries	30	End
29-06-26 23:08:07.873	licationShutdownHook it 'default'	INFO rEntityManagerFactoryBean	destroy	693	Closing JPA EntityManagerFactory for persistence un
29-06-26 23:08:07.875	licationShutdownHook	INFO c.z.h.HikariDataSource	close	359	HikariPool-1 - shutdown initiated...
29-06-26 23:08:07.882	licationShutdownHook	INFO c.z.h.HikariDataSource	close	351	HikariPool-1 - shutdown completed.